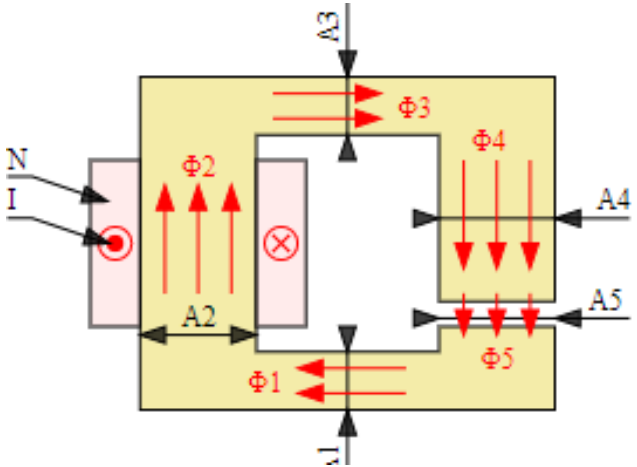
ECE 330

Chapter 4

Magnetic circuits, Mutual Inductance, and Transformers

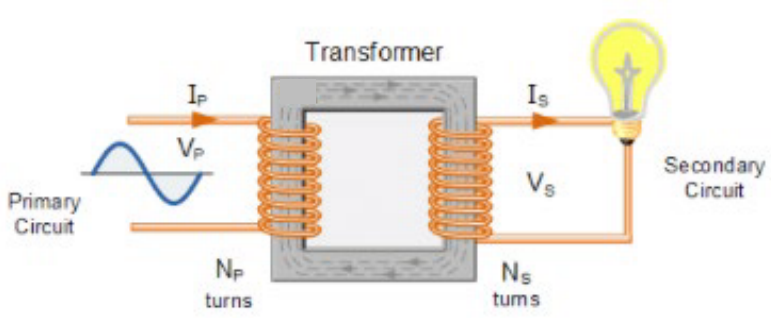


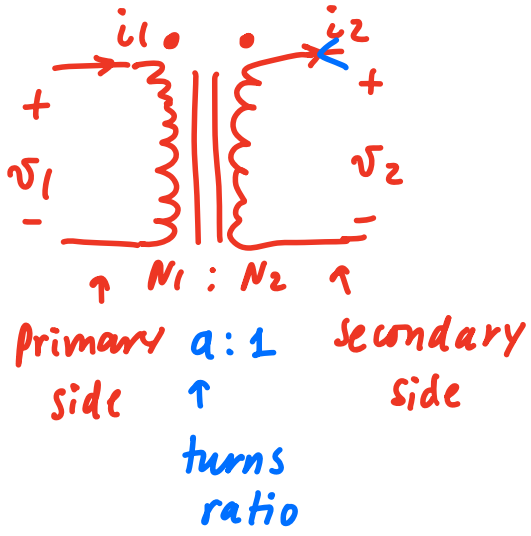
Prof.Olga Mironenko
University of Illinois at Urbana-Champaign
Fall 2026

Learning Objectives: you will be able to

- Analyze **ideal transformers,** and determine voltage, current and impedance relationships between primary and secondary sides.
- Use **Maxwell's equations** to analyze an inductor, and to develop **magnetic equivalent circuits.**
- Model magnetic systems using **magnetic equivalent circuits**, including **reluctance** and **inductance**.
- Evaluate airgap effects, accounting for **airgap reluctance** and **fringing effects.**
- Analyzed **coupled coils**, using **mutual inductance** and **the dot convention.**

Ideal Transformers





$$\frac{v_1}{v_2} = \frac{N_1}{N_2} = a$$

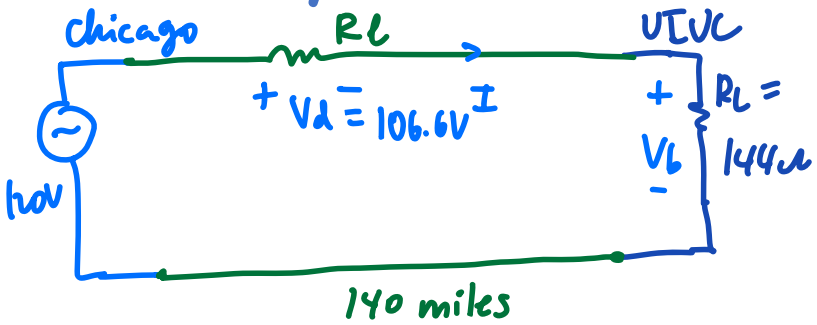
$$\frac{i_1}{-i_2} = \frac{N_2}{N_1} = \frac{1}{a}$$

Q: p_1 and p_2 -?

- $p_1 = v_1 i_1 \quad \frac{N_1}{N_2} \cdot i_1 = a \cdot i_1$
- $p_2 = v_2 i_2 = v_1 i_1 = p_1 \Rightarrow$ Power across ideal transformer is the same.

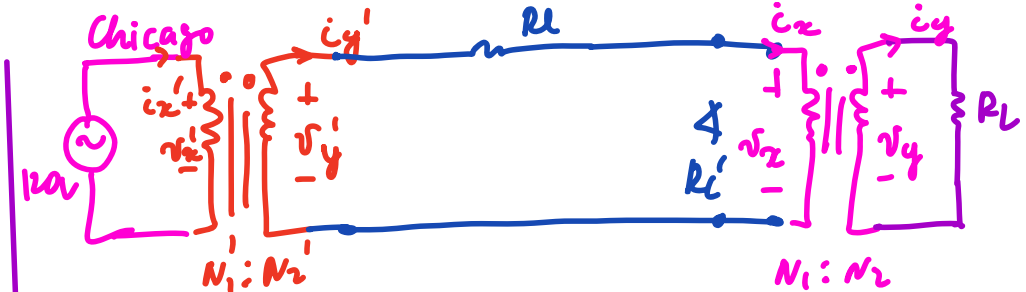
$\uparrow \frac{N_2}{N_1} v_1 = \frac{1}{a} \cdot v_1$

Why Edison lost the War of Currents



$$* R_l = \rho \frac{l}{S}$$

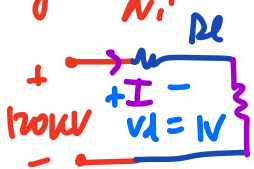
R_l, Ω	I	V_l	η
1148Ω	93 mA	13.4V	11.2%



1 : 1000
step-up transformer

1000 : 1
step-down transformer

$$\bullet\ v_y' = \frac{N_2'}{N_1'} \cdot v_x' = 120kV$$



$$R_l' = (1000)^2 \cdot 144 = 144M\Omega$$

$$I = \frac{120kV}{144M + 1148} = 0.83mA$$

$$\eta = 99.9\%$$

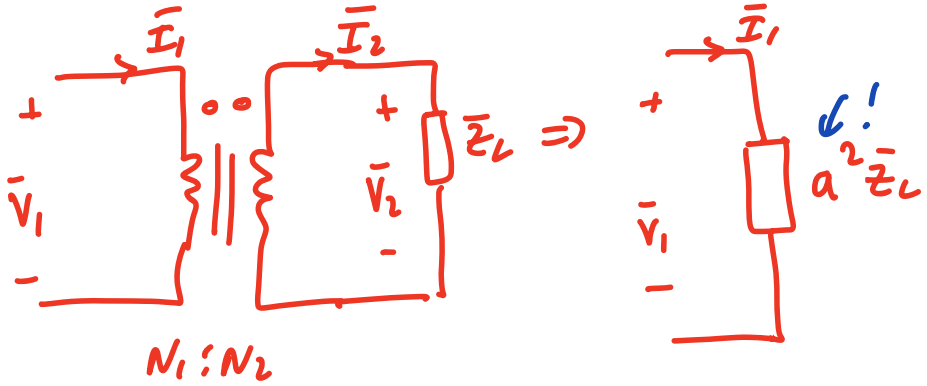
Move impedance across the transformer

$$\bullet\ R_L = \frac{v_y}{i_y} = \frac{N_2 \cdot v_x}{N_1 \left(\frac{N_1}{N_2}\right) i_x}$$

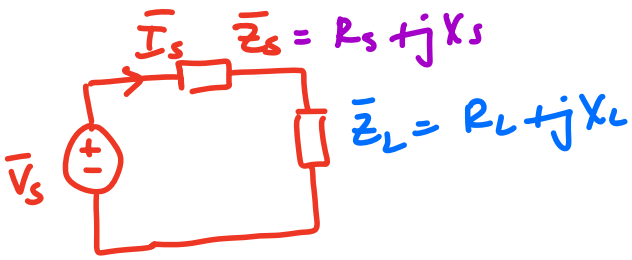
$$\bullet\ \underbrace{\frac{v_x}{i_x}}_{R_l'} = \left(\frac{N_1}{N_2}\right)^2 \cdot R_L = a^2 \cdot R_L$$

$$\boxed{\bar{Z}_p = a^2 \cdot \bar{Z}_s}$$

Moving Impedances Across a Transformer



Maximum Power Transfer



Q: What $\bar{Z}_L$ to choose for max. power transfer?

- $P_L = |\bar{I}_s|^2 \cdot R_L = \dfrac{|\bar{V}_s|^2 \cdot R_L}{(R_s+R_L)^2 + (X_s+X_L)^2}$
- $\bar{I}_s = \dfrac{\bar{V}_s}{\bar{Z}_s+\bar{Z}_L}$

$\Downarrow$ $\quad$ $\Downarrow$

$X_L = -X_s$

$R_L = R_s$

$\Downarrow$

- $X_L = -X_s$
- $R_L = R_s$

$\Downarrow$

$\bar{Z}_L = \bar{Z}_s^*$

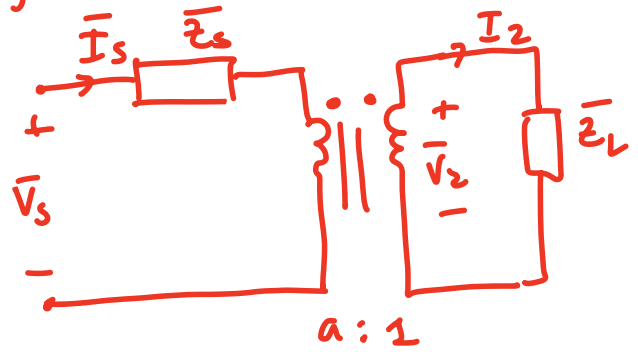
Maximum Power Transfer (Continued)

- Can show for maximum power dissipated at the load
  - The load impedance is the complex conjugate of the line impedance ($\bar{Z}_L = \bar{Z}_s^*$)
  - For a fixed ratio between the resistive and reactive terms of the load impedance, the magnitude of the load impedance is the same as the line impedance

$$|\bar{Z}_L| = |\bar{Z}_s| \text{ for } \frac{R_L}{X_L} = c \text{ (fixed as constant)}$$

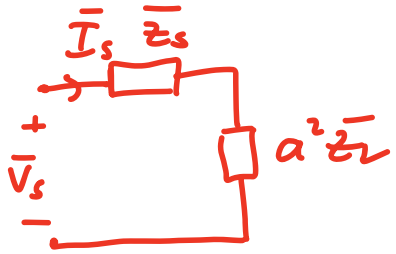
Maximum Power Transfer with Transformer

* Imposing $|\bar{Z}_L| = |\bar{Z}_s|$ can be done with a transformer:



$\Rightarrow$

Moving impedance across the transformer first:

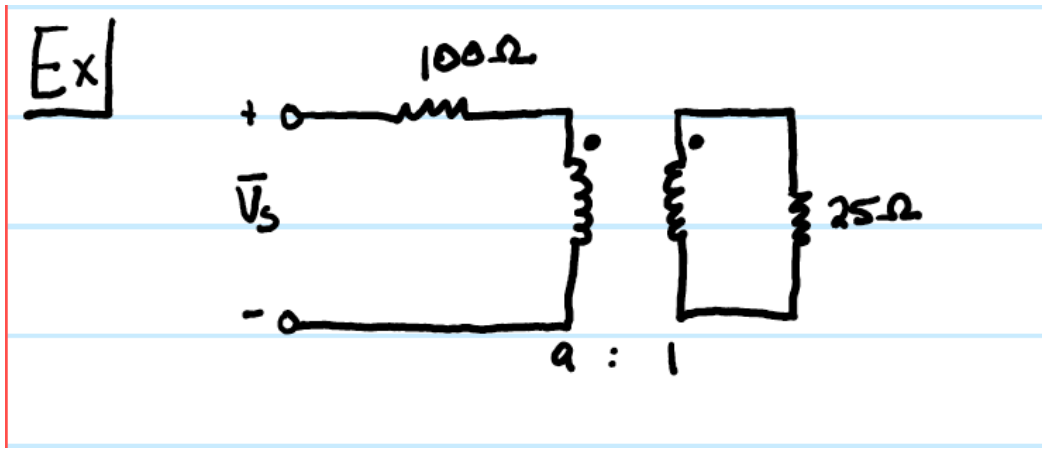


$\Downarrow$

$$|\bar{Z}_s| = |a^2 \bar{Z}_L|$$

can pick "a" for $a = \sqrt{\frac{|\bar{Z}_s|}{|\bar{Z}_L|}}$ a max. power transfer

Example 1



What is *a* for maximum power transfer?

a) 1

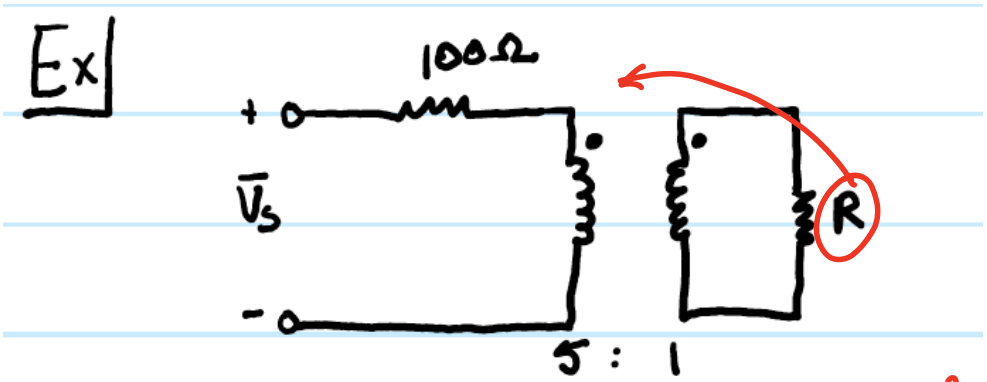
b) 2

c) 3

d) 4

$$a = \sqrt{\frac{R_S}{R_L}} = \sqrt{\frac{100}{25}} = 2$$

Example 2

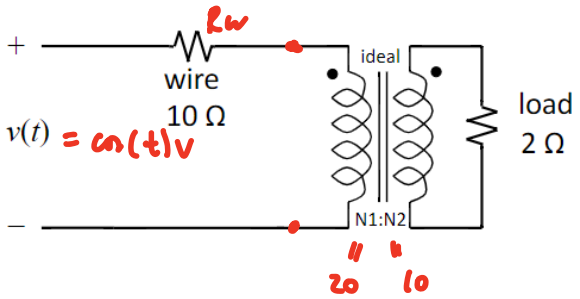


$100 = 5^2 \cdot R \Rightarrow R = 4\Omega$

What is *R* for maximum power transfer?

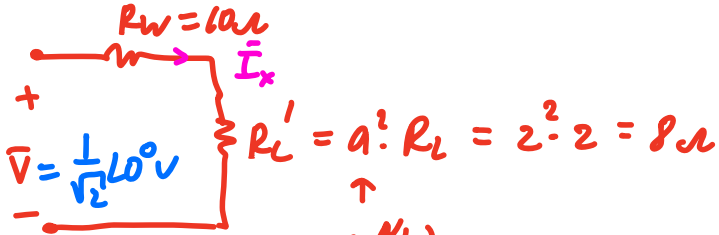
Example 3 (Pr 4d, Spring 2021, M1)

d) The purpose of our transformer is to drive a 2 Ω load from a source that is connected through a 10 Ω wire, as shown below. If v(t) = cos(t) V and we model the transformer as ideal, what is the average power delivered to the load?



① Ideal transf: $P_{in} = P_{out} = ?$

②



$$P_{in} = \bar{I}_x^2 \cdot R_L' = \frac{V^2 \cdot R_L'}{(R_w + R_L')^2} = \frac{1}{81}\,W$$

$\bar{I}_x$ = $\bar{V}$ / ($R_w + R_L'$)

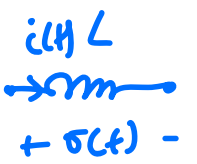
Electromagnetics

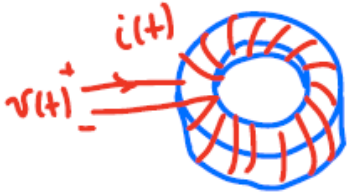
Magnetics: foundation for our electrical equipment: motors, generators, transformers etc.

Key ideas:

① Current produces mag. field
② Changing flux creates a voltage

} ⇒ First, let's start with an inductor

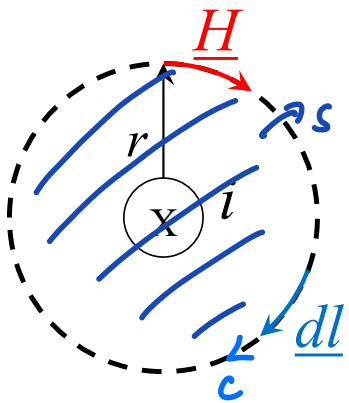




⇒

inject current i(t)
↓
mag. field (intensity), H, [Ampere·turns / meter (A·t / m)]
↓
mag. flux (ϕ), [Wb]
density (B), [Tesla]
↓
changing flux creates a voltage v(t)

Electromagnetics



* current and mag. field

Ampere's law: current density, total current enclosed by s

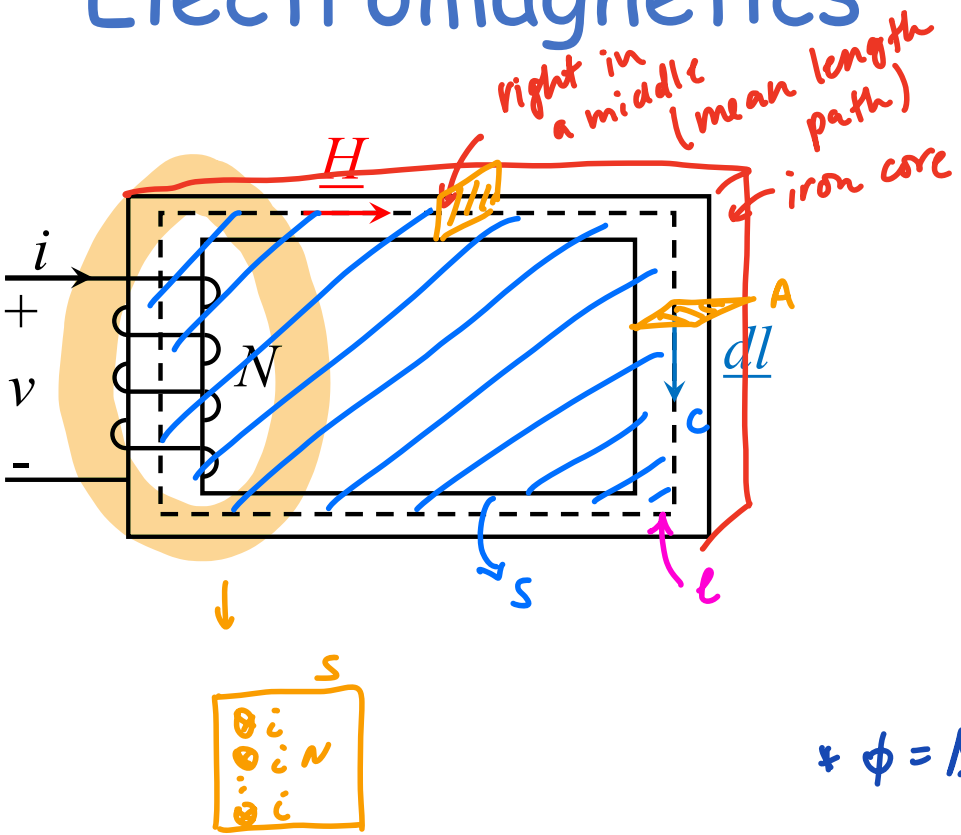
$$\oint_c \underline{H}\cdot \underline{dl} = \int_S \underline{J}_f \cdot \hat{n}\, dA = i$$

surface defined by c

$$\Rightarrow H\cdot \ell = i \qquad H = \frac{i}{2\pi \cdot r}$$

$$\underline{H}\cdot \underline{dl} = H\cdot dl \cdot \underbrace{\cos(\overset{\text{"0"}}{\theta})}_{1} \Rightarrow \oint_c H\,dl = H\cdot \ell$$

Electromagnetics



$$* \oint_c \underline{H} \cdot \underline{dl} = \int_S \underline{J_f} \cdot \hat{n}\, dA = Ni$$

Hl

* Mag. field and mag. flux

$B = \mu H$

magnetic permiability of a material [H/m]

Note: $\mu = \mu_r \mu_0$ ← permiab. of free space: $4\pi \cdot 10^{-7}$ H/m

relative permiability of the material (unitless)

$* \phi = B \cdot A =$

$= \mu H A$

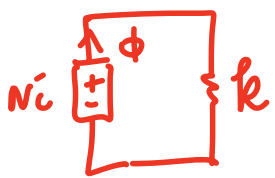
$* \phi = \frac{\mu N i A}{l} = \frac{Ni}{l/\mu A} = \phi$ (voltage / current)

resistance: $R = \frac{l}{\sigma \cdot A}$

Magnetic Equivalent Circuit

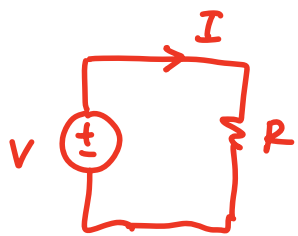
Magnetic

- Magneto-motive force (MMF): Ni [A·t]
- Reluctance $\mathcal{R}$ [$\frac{A \cdot t}{wb}$], [$\frac{1}{H}$]
- mag. flux (ϕ), [Wb]



Electrical

- Voltage : V [V]
- Resistance : R [Ω]
- current : I [A]



What about the $\underline{E}$?

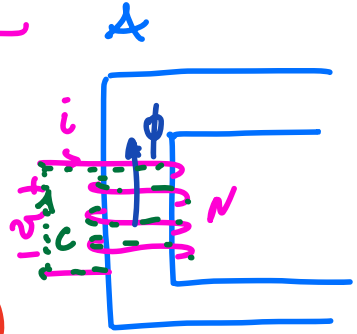
* Changing flux → $v(t)$

Faraday's law:

$$\oint_C \underline{E}\cdot d\underline{l} = -\frac{d}{dt}\left(\int_S \underline{B}\cdot \hat{n}\, dA\right)$$

$$-v + \int_{\text{wire}} \underline{E}\cdot d\underline{l} = -\frac{d}{dt}\, N\cdot B\cdot A \quad (\int \underline{E}\cdot d\underline{l} = 0)$$

$N\cdot B\cdot A = \phi$



Note: assume ideal wire ($\sigma \to \infty$)

$\Rightarrow$ $\underline{E}$ inside the wire is 0

$$v = \frac{d}{dt} N\phi = \frac{d\lambda}{dt}$$

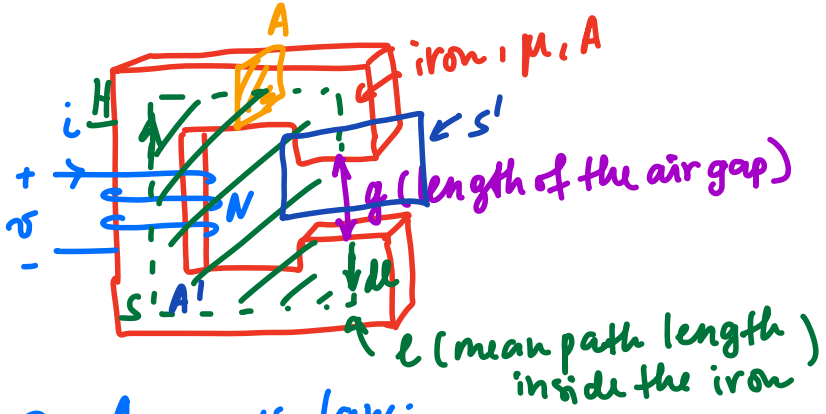
λ: flux linkage (total flux linked by N turns of the coil)

Inductance

- $\phi = \dfrac{Ni}{\ell/\mu A} \longrightarrow$ $\lambda = N\phi = \underbrace{\dfrac{N^2 \mu A}{\ell}}_{L\,[H]} \cdot i \Rightarrow$ $\boxed{\lambda = L \cdot i}$

$$\boxed{v = \frac{d\lambda}{dt} = L\frac{di}{dt}}$$

Iron Core with Air Gap



① Ampere's law:

$$\oint_C \underline{H}\cdot d\underline{l} = \int_S \underline{J_f}\cdot\hat{n}\, dA'$$

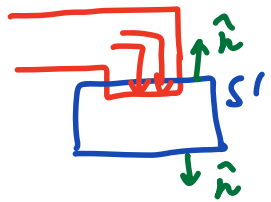
$$\Downarrow$$

$$H_{iron}\cdot l + H_{gap}\cdot g = Ni$$

② Gauss's law: conservation of flux

$$\oint_{s'} \underline{B}\cdot\hat{n}\, dA = 0$$

↑ closed surface s'



$B_g \cdot A_g - B_{iron}\cdot A_{iron} = 0$

$B_g \cdot A_g$: flux flows out

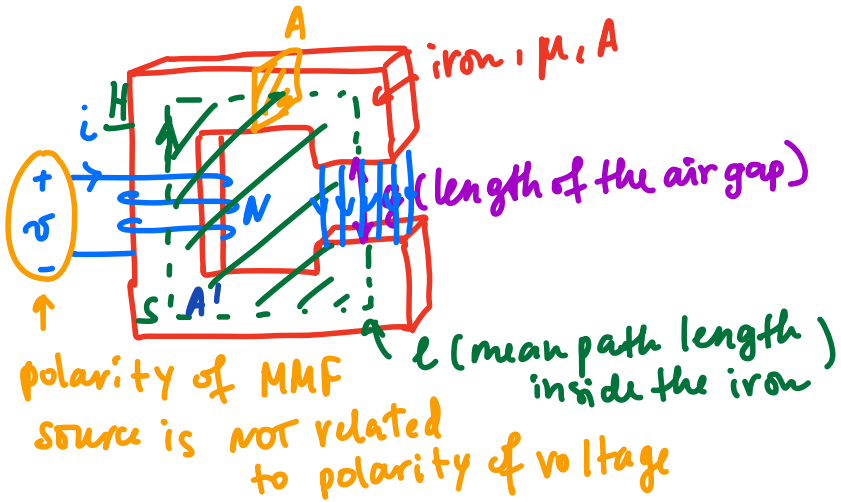
$B_{iron}\cdot A_{iron}$: flux flows in

Note: neglect fringing ($A_g = A_{iron}$)

$$\Downarrow$$

$$B_g = B_{iron}$$

Iron Core with Air Gap



③ $B_{iron} = \mu H_{iron}$

$B_{gap} = \mu_r \mu_0 H_g$

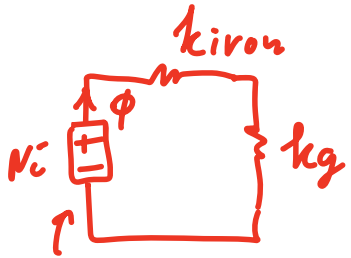
$\mu_r = 1$ (air)

$H_{iron} \cdot \ell + H_{gap} \cdot g = Ni \Rightarrow \frac{B_{iron} \cdot \ell}{\mu} + \frac{B_g \cdot g}{\mu_0} = Ni$

④ $\phi = B_{iron} \cdot A_{iron} = B_g \cdot A_g \Rightarrow B_{iron} = \frac{\phi}{A_{iron}}$

$B_g = \frac{\phi}{A_g}$

⑤ $\phi \left(\underbrace{\frac{\ell}{A_{iron} \cdot \mu}}_{k_{iron}} + \underbrace{\frac{g}{A_g \cdot \mu_0}}_{k_g} \right) = Ni$



polarity: ϕ flows out of "+" into "−"

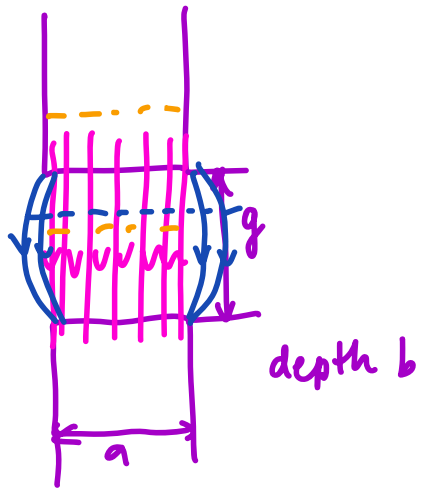
Back to

Faraday's Law

- $v = \frac{d\lambda}{dt}$ where $\lambda = N\phi = \underbrace{\frac{N^2}{\mathcal{R}_{iron} + \mathcal{R}_g}}_{L} \cdot i = L \cdot i$

Note: $\mathcal{R}_{iron} = \frac{\ell}{\mu A_{iron}}$; if $\mu \to \infty$, $\mathcal{R}_{iron} \to 0$

Fringing Effects



* No fringing : $A_g = a \cdot b$

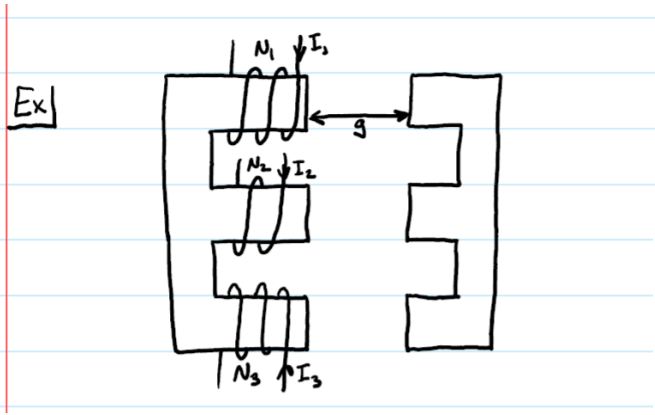
* Fringing : $A_g = (a+g)(b+g)$

Note: g ↓, fringing effects are reduced

Q: $k = \frac{l}{\mu A} \Rightarrow k_g = \frac{g}{\mu_0 A_g}$ (k_g ↓, A_g ↑)

If we consider fringing, what happens to k_g? Decrease

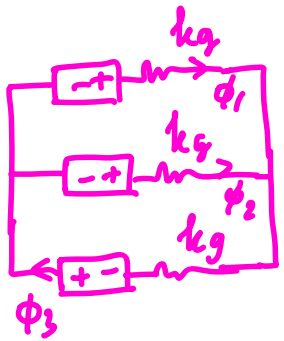
Example 4



Given: $N_1 = 100$ $I_1 = 25A$ $A_g = 4cm^2$

$N_2 = 50$ $I_2 = 10A$ $\mu = \infty$ → $k_{iron} = 0$

$N_3 = 100$ $I_3 = 15A$ $g = 0.1cm$



Where should the positive terminals be placed on the MMF sources?

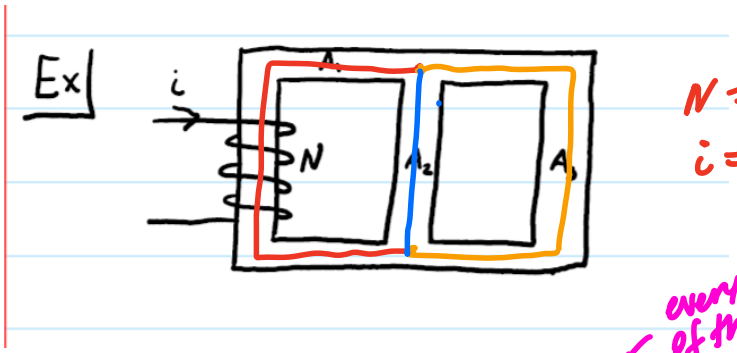
a) right, left, right

b) left, left, right

c) right, right, left

d) right, right, right

Example 5



$N = 25$

$i = 0.5\,A$

$\mu_{r1} = 2250 = \mu_{r3}$

$l_1 = 30\,cm = l_3$ (cm → m)

$A_1 = 2\,cm^2 = A_3$ (cm^2 → m^2)

$\mu_{r2} = 1350$

$l_2 = 10\,cm$

$A_2 = 4\,cm^2$

Q: find flux density B_1, B_2, B_3 [T]

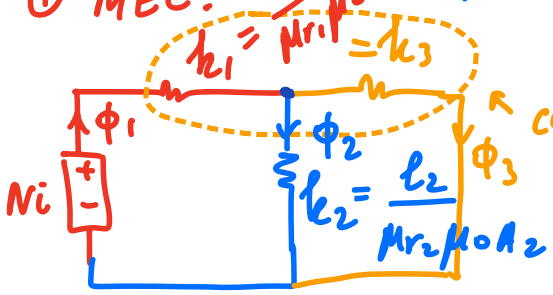
$$B_1 = \frac{\Phi_1}{A_1} = 0.0966\,T$$

$$B_2 = \frac{\Phi_2}{A_2} = 0.0377\,T$$

$$B_3 = \frac{\Phi_3}{A_3} = 0.021\,T$$

① MEC: $\mathcal{R}_1 = \frac{l_1}{\mu_{r1}\mu_0 A_1} = \mathcal{R}_3$

Note: $\mathcal{R} = \frac{l}{\mu_r \mu_0 A}$ ← every time any of those change, we need to introduce new reluctance



↖ can't combine them, because they are not in series

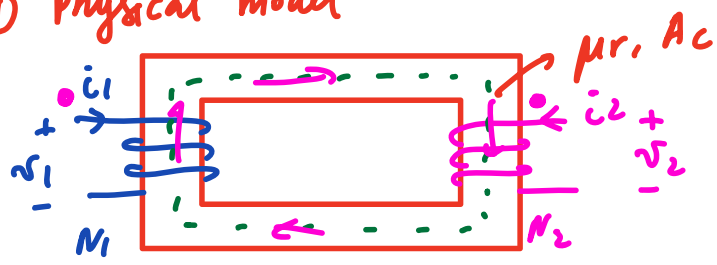
② Solve MEC:

- $\Phi_1 = \frac{Ni}{\mathcal{R}_{tot}} = \frac{Ni}{\mathcal{R}_1 + \left(\frac{\mathcal{R}_2 \cdot \mathcal{R}_3}{\mathcal{R}_2 + \mathcal{R}_3}\right)} = 19.32 \cdot 10^{-6}\,Wb = 6.46 \cdot 10^5 \frac{At}{m}$
- $\Phi_2 = \Phi_1 \left(\frac{\mathcal{R}_3}{\mathcal{R}_2 + \mathcal{R}_3}\right) = 15.11 \cdot 10^{-6}\,Wb$
- $\Phi_3 = \Phi_1 - \Phi_2 = 4.21 \cdot 10^{-6}\,Wb$

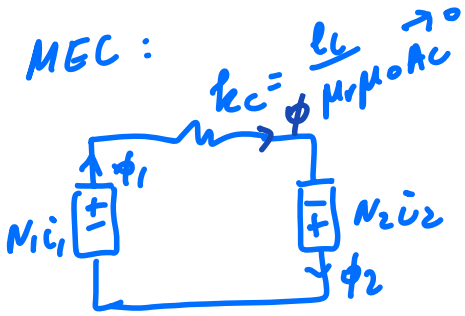
Coupled Coils: ideal case

Assume: 1) $\mu_c = \infty$
2) Lossless coils
3) Perfect coupling
all flux that links coil 1 also links coil 2.

① Physical model



② MEC:



$N_1 i_1 + N_2 i_2 = 0$

$N_1 i_1 = -N_2 i_2$

$$\frac{i_1}{-i_2} = \frac{N_2}{N_1} \quad (*)$$

③ Now voltages:

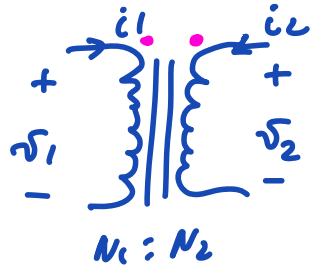
$$v_1 = \frac{d\lambda_1}{dt} = \frac{d}{dt}(N_1\phi) = N_1 \frac{d\phi}{dt}$$

$$v_2 = \frac{d\lambda_2}{dt} = \frac{d}{dt}(N_2\phi) = N_2 \frac{d\phi}{dt}$$

$$\boxed{\frac{v_1}{v_2} = \frac{N_1}{N_2}} \quad (*)$$

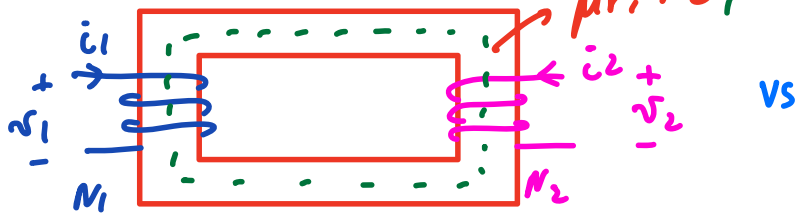
, Mathematical model"

④ Electrical circuit model:

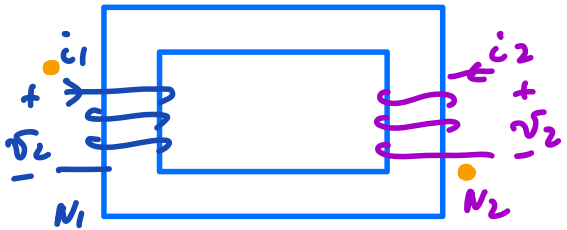


Dot convention: we want our reference model to be valid irrespective of winding direction.

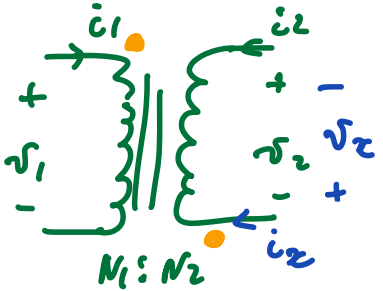
① Physical model



vs

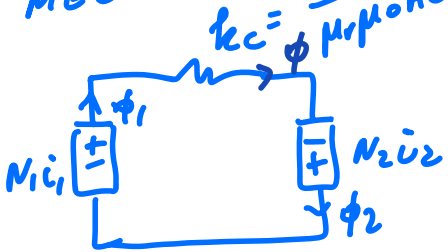


OR

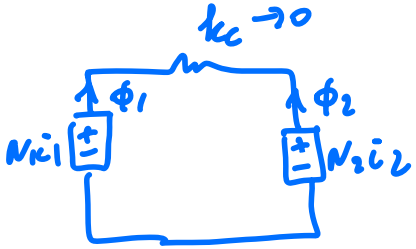


② MEC:

$$\mathcal{R}_c = \frac{l_c}{\mu_r \mu_0 A_c} \to 0$$



$$N_1 i_1 + N_2 i_2 = 0$$



$$N_1 i_1 - N_2 i_2 = 0$$

MEC is now different! which means we have to know how coils are physically wound ∴

$$i_x = -i_2$$

$$v_x = -v_2$$

Ref: $\frac{i_1}{-i_x} = \frac{N_2}{N_1}$ $\frac{v_1}{v_x} = \frac{N_1}{N_2}$

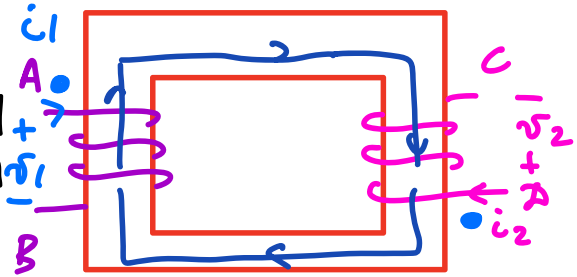
⇓

Substitute

$$\boxed{\frac{i_1}{i_2} = \frac{N_2}{N_1} \quad \frac{v_1}{-v_2} = \frac{N_1}{N_2}}$$

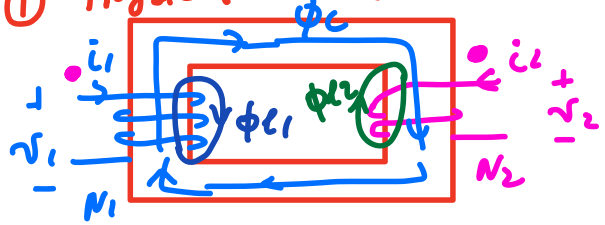
Dot convention: "currents entering dots throw flux in the same direction"

- Step 1) Select a coil and put a dot on one terminal- arbitrarily (your choice)
- Step 2) Mark the dotted terminal with + voltage polarity and inject current into first dot
- Step 3) Use RHR to determine flux direction
- Step 4) Determine current direction in the second winding which will throw flux in the same direction as the first winding
- Step 5) Put the dot in the second winding, where current enters the terminal
- Step 6) Put + voltage polarity on the second dotted terminal



Coupled Coils: practical case

① Physical model



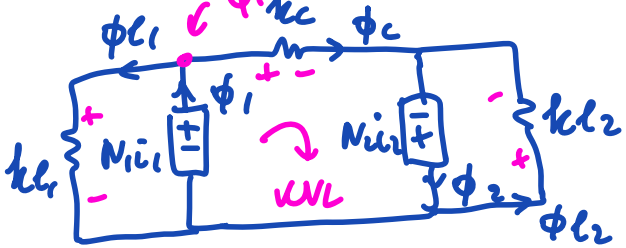
↳ $\mu_c \neq \infty$

ϕ_c : flux linking both coils

ϕ_{ℓ_1} : flux linking coil 1 but not coil 2

ϕ_{ℓ_2} : flux linking coil 2 but not coil 1

② MEC: $\phi_1 = \phi_c + \phi_{\ell_1}$



③ Now, find voltages:

$$v_1 = \frac{d\lambda_1}{dt} = \frac{d}{dt}(N_1\phi_1)$$

$$\lambda_1 = N_1\phi_1 = N_1\left(\frac{N_1 i_1}{\mathcal{R}_{\ell_1}} + \frac{N_1 i_1 + N_2 i_2}{\mathcal{R}_c}\right) =$$

$$= \underbrace{\left[\frac{N_1^2}{\mathcal{R}_c} + \frac{N_1^2}{\mathcal{R}_{\ell_1}}\right]}_{L_1} i_1 + \underbrace{\frac{N_1 N_2}{\mathcal{R}_c}}_{M} i_2$$

L_1 - self-inductance of coil 1

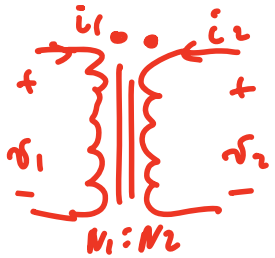
M - mutual inductance between coil 1 and coil 2

$$v_2 = \frac{d\lambda_2}{dt} = \frac{d}{dt}(N_2\phi_2)$$

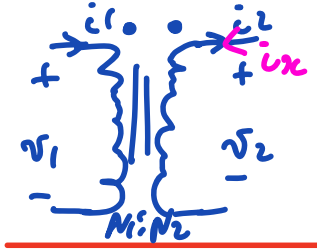
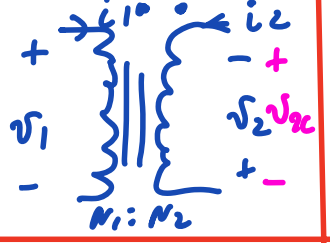
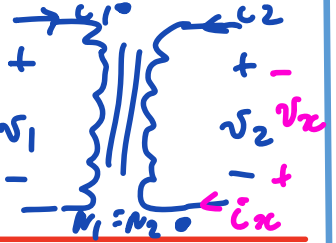
$$\vdots$$

$$\lambda_2 = \underbrace{\left[\frac{N_1 N_2}{\mathcal{R}_c}\right]}_{M} i_1 + \underbrace{\left[\frac{N_2^2}{\mathcal{R}_{\ell_2}} + \frac{N_2^2}{\mathcal{R}_c}\right]}_{L_2} i_2$$

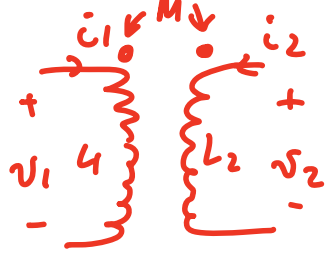
Ideal:



$$\frac{i_1}{-i_2} = \frac{N_2}{N_1} \qquad \frac{v_1}{v_2} = \frac{N_1}{N_2}$$

If	If	If
		
$i_x = -i_2$	$v_x = -v_2$	$i_x = -i_2$ $v_x = -v_2$
Ref: $\frac{i_1}{-i_x} = \frac{N_2}{N_1}$ ⇓ sub. $\frac{i_1}{i_2} = \frac{N_2}{N_1}$ $\frac{v_1}{v_2} = \frac{N_1}{N_2}$	Ref: $\frac{v_1}{v_x} = \frac{N_1}{N_2}$ ⇓ sub. $\frac{i_1}{-i_2} = \frac{N_2}{N_1}$ $\frac{v_1}{-v_2} = \frac{N_1}{N_2}$	Ref: $\frac{i_1}{-i_x} = \frac{N_2}{N_1}$ $\frac{v_1}{v_x} = \frac{N_1}{N_2}$ ⇓ sub $\frac{i_1}{i_2} = \frac{N_2}{N_1}$ $\frac{v_1}{-v_2} = \frac{N_1}{N_2}$

Practical:

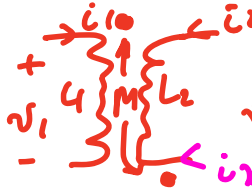


$$\lambda_1 = L_1 i_1 + M i_2$$
$$\lambda_2 = M i_1 + L_2 i_2$$

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$
$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

Note: L_1, L_2, M are never negative!

Ex:



v_1 - ?

v_2 - ?

Ref: $v_1 = L_1 \frac{di_1}{dt} + M \frac{di_x}{dt}$

$v_x = M \frac{di_1}{dt} + L_2 \frac{di_x}{dt}$

$v_x = -v_2$

$i_x = -i_2$ ⇒ sub:

$$v_1 = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$
$$v_2 = -M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

Coefficient of Coupling

$$K = \frac{M}{\sqrt{L_1 L_2}}$$

$0 \le K \le 1$

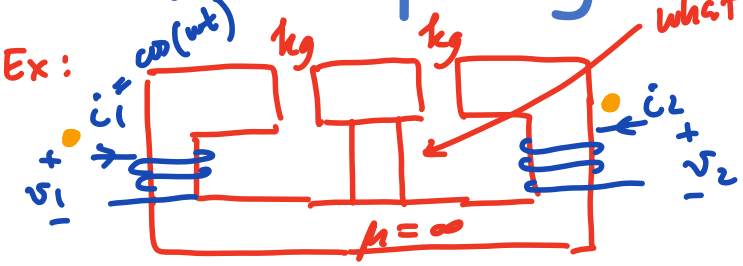
↑ no coupling ↑ perfect coupling

Side note:

$\frac{di_1}{dt} \rightarrow j\omega \bar{I}_1$

$\omega = 2\pi f$

Ex: $\cos(\omega t)$

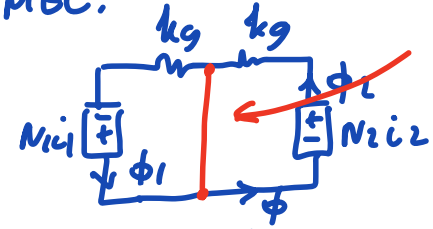


what if we insert the middle segment with $\mu = \infty$?

$v_1 - ?$

$v_2 - ?$

① MEC:



due to the middle segment:

$\phi_1 = \frac{N_1 i_1}{kg}$; $\phi_2 = \frac{N_2 i_2}{kg}$

$\phi_1 = \phi_2 = \phi$

No M!

$\bar{V}_1 = j\omega L_1 \bar{I}_1$

$\bar{V}_2 = j\omega M \bar{I}_1 = 0$ $(K=0)$

② $\lambda_1 = N_1\phi_1 = \frac{N_1^2}{2kg} i_1 + \frac{N_2 N_1}{2kg} i_2$ (L_1, M)

$\lambda_2 = N_2\phi_2 = \frac{N_1 N_2}{2kg} i_1 + \frac{N_2^2}{2kg} i_2$ (M, L_2)

③ Check dots:

$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$

$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$

← Now, let's say, 2nd coil is open $\Rightarrow i_2 = 0$

$\Rightarrow \bar{V}_1 = j\omega \bar{I}_1 L_1$

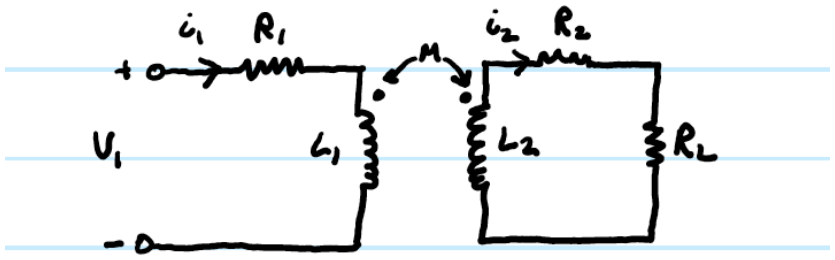
$\bar{V}_2 = j\omega M \bar{I}_1$

$\Rightarrow \frac{\bar{V}_1}{\bar{V}_2} = \frac{M}{L_1} = \frac{N_2}{N_1}$ $(K=1)$

ideal transf.

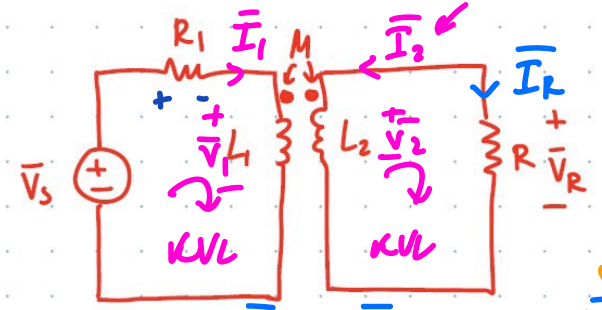
Example 6

Real Transformers



- Real transformers have resistance in the wire, inductance from the core, and mutual inductance between the two windings.

assign labels on coupled coils so that it matches ref.



Given: $\omega, \bar{V}_S, R_1, R, L_1, L_2, M$

Find: $\bar{V}_R$

Approach: ① $\bar{V}_R = \bar{I}_R \cdot R = -\bar{I}_2 \cdot R$

② KVL: $-V_S + \bar{I}_1 \cdot R_1 + \bar{V}_1 = 0$

KVL: $-\bar{V}_2 + (-\bar{I}_2 \cdot R) = 0$

Ref: $\bar{V}_1 = j\omega \bar{I}_1 L_1 + j\omega \bar{I}_2 M$

$\bar{V}_2 = j\omega M L_1 + j\omega \bar{I}_2 L_2$

③ $\bar{I}_1 \cdot R_1 + j\omega \bar{I}_1 L_1 + j\omega \bar{I}_2 M = V_S$

$\bar{I}_2 \cdot R + j\omega M \bar{I}_1 + j\omega \bar{I}_2 L_2 = 0$

} solve for $\bar{I}_2$

Side note: Thevenin Equivalent (not on quizzes/exams)

